

Same-species ancilla interaction in the three-qubit gates

Adding the Cs–Cs van der Waals coupling V_{cc} to the OR and CCX average-fidelity simulations

TriQG examples/Average_fidelity_Vcc · compiled 2026-04-11

TL;DR. The previous `Average_fidelity` simulations only carried the data–ancilla blockade V_{ct} and dropped the same-species ancilla–ancilla coupling V_{cc} entirely ($V_{cc} = 0$). The SelfCorrectingRydberg paper (main.tex, Table I) specifies a Cs–Cs van der Waals coefficient $C_6 = -1449 \text{ GHz}\cdot\mu\text{m}^6$ at the rotated lattice spacing $r_{AA} = a\sqrt{2} \approx 7.07 \text{ }\mu\text{m}$, which gives $|V_{cc}|/(2\pi) \approx 11.6 \text{ MHz}$. With this term switched on:

- OR gate: $\overline{F}_{\text{OR}} = 0.996688 \rightarrow 0.992597$, infidelity $3.31 \times 10^{-3} \rightarrow 7.40 \times 10^{-3}$.
- CCX gate: $\overline{F}_{\text{CCX}} = 0.999602 \rightarrow 0.997331$, infidelity $3.98 \times 10^{-4} \rightarrow 2.67 \times 10^{-3}$.

The impact is strongly branch-selective: only the inputs that drive both Cs controls into $|r\rangle$ simultaneously are affected, and the size of the drop is consistent with the ratio $|V_{cc}|/\Omega_c \approx 0.23$ detuning the joint $|rr\rangle$ de-excitation π -pulse.

1 Motivation

The three-qubit OR and CCX gates in this protocol use two same-species control atoms (both ^{133}Cs in the paper’s worked example) and one target atom (^{87}Rb). During each gate, the control atoms can be temporarily excited to the Rydberg state $|r\rangle = |79D_{5/2}\rangle$, where they interact with each other through van der Waals forces just as they interact with the target through the stronger Rb–Cs dipole-dipole Förster coupling. In the previous `Average_fidelity` scripts the ancilla–ancilla interaction was dropped entirely – only the data–ancilla V_{ct} term appeared in the Hamiltonian. That omission hides a real physical error channel that becomes visible whenever two Cs atoms are promoted to $|r\rangle$ at the same time.

This report documents the effect of reinstating the Cs–Cs interaction V_{cc} with the value quoted in main.tex, Table I.

2 Physical coefficient

From main.tex, Table I (Rydberg interaction parameters at lattice spacing $a = 5 \text{ }\mu\text{m}$):

Quantity	Symbol	Value
Pair	Cs–Cs (A_1 – A_1)	$ 79D_{5/2}\rangle + 79D_{5/2}\rangle$
van der Waals coefficient	C_6	$-1449 \text{ GHz}\cdot\mu\text{m}^6$
Rotated lattice spacing	a	$5 \text{ }\mu\text{m}$
Nearest same-type ancilla distance	$r_{AA} = a\sqrt{2}$	$5\sqrt{2} \approx 7.07 \text{ }\mu\text{m}$
Interaction strength (magnitude)	$ V_{cc} /(2\pi)$	$\approx 11.6 \text{ MHz}$
Signed interaction (used here)	$V_{cc}/(2\pi)$	-11.6 MHz

Table 1: Cs–Cs (A_1 – A_1) same-species interaction parameters from main.tex, Table I. The paper quotes the magnitude; we use the signed value in the simulation (see §3.2).

The numerical check:

$$V_{cc} = \frac{C_6}{r_{AA}^6} = \frac{-1449 \text{ GHz} \cdot \mu\text{m}^6}{(5\sqrt{2})^6 \mu\text{m}^6} = \frac{-1449}{125\,000} \text{ GHz} \approx -11.59 \text{ MHz},$$

consistent with the paper’s quoted magnitude of 11.6 MHz.

For reference, the same-round data-ancilla blockade already present in the scripts is the Rb-Cs dipole-dipole Förster coupling $V_{ct}/(2\pi) = 593 \text{ MHz}$ (Table I, Rb-Cs row), so the ratio $V_{ct}/|V_{cc}| \approx 51$ matches the paper’s “selectivity” figure.

3 Implementation

3.1 Hamiltonian changes

Both builders in `triqq/hamiltonian.py` now accept an optional `V_cc` argument defaulting to 0.0, so existing callers (including the scripts in `examples/Average_fidelity/`) are unaffected. When $V_{cc} \neq 0$ the following static term is added to `H_static`:

$$H_{cc} = V_{cc} |r\rangle\langle r|_{c_1} \otimes |r\rangle\langle r|_{c_2} \otimes \mathbb{1}_t.$$

In code this is a single extra line:

```
# Same-species control-control van der Waals interaction:
# V_cc * |r><r|_c1 @ |r><r|_c2 @ I_t (main.tex, Table I, Cs-Cs row)
H_ancilla_ancilla = V_cc * proj_r_c1 * proj_r_c2
```

This is a diagonal operator that only gives an energy shift to the $|rr\rangle$ branch of the three-atom Hilbert space; it does not introduce any new couplings or new decay channels.

3.2 Sign convention

The Table I column lists the **magnitude** $|V_{cc}|/(2\pi) = 11.6 \text{ MHz}$, but the underlying coefficient $C_6 = -1449 \text{ GHz} \cdot \mu\text{m}^6$ is negative, so the physical interaction energy $V_{cc} = C_6/r_{AA}^6$ is also negative. We pass the signed value $V_{cc} = -2\pi \times 11.6 \text{ MHz}$ to the builder. For the current simulation – average gate fidelity over all computational-basis inputs – the sign is a global phase on the $|rr\rangle$ branch and has no effect on any F_k . But using the signed value keeps the Hamiltonian physically faithful and makes the numbers correct for any downstream use with superposition inputs or multi-gate sequences.

3.3 Script changes

Both scripts in `examples/Average_fidelity_Vcc/` now:

1. declare `V_cc = -2*np.pi * 11.6 # Cs-Cs vdW at r_AA = 5*sqrt(2) um,`
2. pass `V_cc=V_cc` into `build_hamiltonian(...)` (OR) or `build_ccx_hamiltonian(...)` (CCX),
3. print both V_{ct} and V_{cc} in the run header for traceability.

Nothing else is modified – pulse amplitudes, detunings, timing windows, decoherence rates, and all solver options remain exactly the same as the paper-consistent run documented in `results/Final_three_qubit_gate_report.typ`.

4 OR gate results

The OR gate excites controls on the $|1\rangle \rightarrow |r\rangle$ branch, so the $|1, 1, \ast\rangle$ inputs are the ones that put both Cs atoms in $|r\rangle$ simultaneously during the long ($\approx 300 \text{ ns}$) target pulse.

#	Input	Ideal output	F_k , no V_{cc}	F_k , with V_{cc}
1	$ 0, 0, A\rangle$	$ 0, 0, A\rangle$	0.999186	0.999186
2	$ 0, 0, B\rangle$	$ 0, 0, B\rangle$	0.999186	0.999186
3	$ 0, 1, A\rangle$	$ 0, 1, B\rangle$	0.995761	0.995761
4	$ 0, 1, B\rangle$	$ 0, 1, A\rangle$	0.995761	0.995761
5	$ 1, 0, A\rangle$	$ 1, 0, B\rangle$	0.995761	0.995761
6	$ 1, 0, B\rangle$	$ 1, 0, A\rangle$	0.995761	0.995761
7	$ 1, 1, A\rangle$	$ 1, 1, B\rangle$	0.996042	0.979678
8	$ 1, 1, B\rangle$	$ 1, 1, A\rangle$	0.996042	0.979678

Table 2: OR gate: per-input state fidelities with and without the Cs–Cs interaction. The $|0, 0, *\rangle$, $|0, 1, *\rangle$, and $|1, 0, *\rangle$ branches are numerically identical because they place at most one Cs in $|r\rangle$, so the V_{cc} term has no matrix element on them.

OR gate average fidelity: $\bar{F}_{\text{OR}} = 0.992597$ (vs. 0.996688 without V_{cc}), infidelity $1 - \bar{F}_{\text{OR}} = 7.40 \times 10^{-3}$ (vs. 3.31×10^{-3}).

The output-population breakdown confirms the branch-selectivity. Only the $|1, 1, *\rangle$ target populations change; all others are bit-for-bit identical to the no- V_{cc} run.

Input	$P(A)$	$P(B)$	$P(P)$	$P(R)$
$ 0, 0, A\rangle$	0.9992	0.0006	0.0000	0.0002
$ 0, 0, B\rangle$	0.0006	0.9992	0.0000	0.0002
$ 0, 1, A\rangle$	0.0033	0.9958	0.0000	0.0000
$ 0, 1, B\rangle$	0.9958	0.0033	0.0000	0.0000
$ 1, 0, A\rangle$	0.0033	0.9958	0.0000	0.0000
$ 1, 0, B\rangle$	0.9958	0.0033	0.0000	0.0000
$ 1, 1, A\rangle$	0.0023	0.9797	0.0000	0.0000
$ 1, 1, B\rangle$	0.9797	0.0023	0.0000	0.0000

Table 3: OR gate: target-level populations after the gate, with V_{cc} switched on. Highlighted rows are the $|1, 1, *\rangle$ inputs where both Cs controls are in $|r\rangle$ during the target pulse. Compare with `results/Final_three_qubit_gate_report.typ` Table 4 for the $V_{cc} = 0$ baseline; all other rows are unchanged.

The $|1, 1, *\rangle$ branch drops from ≈ 0.9960 to ≈ 0.9797 , i.e.

about 1.6% of population is lost from the intended flipped target state. Leakage to $|P\rangle$ and $|R\rangle$ remains zero; the missing population is in states outside the target Hilbert space, which corresponds to residual Cs excitation – control atoms that were supposed to return to $|1\rangle$ but are still trapped in $|r\rangle$ at the end of the gate (see §6 for the mechanism).

5 CCX gate results

The CCX gate excites controls on the $|0\rangle \rightarrow |r\rangle$ branch (opposite convention), so the $|0, 0, *\rangle$ inputs are the ones that put both Cs atoms in $|r\rangle$ during the three resonant target sub-pulses.

#	Input	Ideal output	F_k , no V_{cc}	F_k , with V_{cc}
1	$ 0, 0, A\rangle$	$ 0, 0, A\rangle$	0.999512	0.990583
2	$ 0, 0, B\rangle$	$ 0, 0, B\rangle$	0.999282	0.990044
3	$ 0, 1, A\rangle$	$ 0, 1, A\rangle$	0.999736	0.999736
4	$ 0, 1, B\rangle$	$ 0, 1, B\rangle$	0.999307	0.999307
5	$ 1, 0, A\rangle$	$ 1, 0, A\rangle$	0.999736	0.999736
6	$ 1, 0, B\rangle$	$ 1, 0, B\rangle$	0.999307	0.999307
7	$ 1, 1, A\rangle$	$ 1, 1, B\rangle$	0.999968	0.999968
8	$ 1, 1, B\rangle$	$ 1, 1, A\rangle$	0.999968	0.999968

Table 4: CCX gate: per-input state fidelities with and without the Cs–Cs interaction. The $|0, 1, *\rangle$, $|1, 0, *\rangle$, and $|1, 1, *\rangle$ branches are numerically identical. Only $|0, 0, *\rangle$ changes – mirror image of the OR pattern, reflecting the opposite excitation convention.

CCX gate average fidelity: $\bar{F}_{\text{CCX}} = 0.997331$ (vs. 0.999602 without V_{cc}), infidelity $1 - \bar{F}_{\text{CCX}} = 2.67 \times 10^{-3}$ (vs. 3.98×10^{-4}).

The CCX relative degradation is larger than the OR relative degradation ($6.7 \times$ vs. $2.2 \times$ increase in infidelity), even though the per-input drop is smaller in absolute terms ($\approx 0.9\%$ for the two affected CCX inputs vs. $\approx 1.6\%$ for the OR). The reason is simple arithmetic: the baseline CCX infidelity is much lower (3.98×10^{-4}), so the fixed absolute penalty of $\approx 9 \times 10^{-3}$ on two input branches dominates the average.

6 Summary comparison

Metric	No V_{cc}	With $V_{cc} = -2\pi \times 11.6$ MHz
OR gate $\bar{F}$	0.996688	0.992597
OR gate $1 - \bar{F}$	3.31×10^{-3}	7.40×10^{-3}
OR affected inputs	–	$ 1, 1, A\rangle, 1, 1, B\rangle$
OR per-input drop	–	$\approx 1.64\%$
CCX gate $\bar{F}$	0.999602	0.997331
CCX gate $1 - \bar{F}$	3.98×10^{-4}	2.67×10^{-3}
CCX affected inputs	–	$ 0, 0, A\rangle, 0, 0, B\rangle$
CCX per-input drop	–	$\approx 0.9\%$
OR gate time	320 ns	320 ns
CCX gate time	40 ns	40 ns

Table 5: Summary: paper-consistent run without and with the Cs–Cs ancilla–ancilla interaction. Gate timings, pulse amplitudes, detunings, and decoherence rates are identical in both columns – the only change is $V_{cc} = 0$ vs. $V_{cc} = -2\pi \times 11.6$ MHz.

Two observations.

1. **Strict branch selectivity.** The V_{cc} term is a diagonal shift on $|rr*\rangle$, so it only couples into the dynamics when both Cs atoms are actually driven to $|r\rangle$ together. Every other input branch is unchanged at the 10^{-6} level. This is a clean stress test that the new term is wired in correctly.
2. **CCX and OR are affected on opposite halves of the truth table.** The OR gate uses the $|1\rangle \rightarrow |r\rangle$ convention and sees V_{cc} on $|1, 1, *\rangle$; the CCX gate uses the opposite $|0\rangle \rightarrow |r\rangle$ convention

and sees V_{cc} on $|0,0,*\rangle$. This mirror symmetry in the data is a direct consequence of the paper’s choice (main.tex §III.B) to use opposite control-pulse polarities for the two gates and is a further consistency check on the implementation.

7 Why the drop is the size it is (qualitative)

The physical picture is a textbook detuned-Rabi calculation. In the two-control subspace

$$\{|11\rangle, |r1\rangle, |1r\rangle, |rr\rangle\},$$

the Cs drive Ω_c couples neighboring states and V_{cc} appears only as a diagonal shift on the corner $|rr\rangle$. For the OR gate’s excitation π -pulse starting from $|11\rangle$:

- The first photon transfers $|11\rangle \rightarrow (|r1\rangle + |1r\rangle)/\sqrt{2}$ resonantly.
- The second photon attempts $|r1\rangle \rightarrow |rr\rangle$ (or the symmetric $|1r\rangle \rightarrow |rr\rangle$), but now the target state sits at energy V_{cc} , so the transfer is **detuned**.

For a two-level Rabi drive with Rabi frequency Ω_c and detuning $\Delta = V_{cc}$, the maximum transfer probability is

$$P_{\max} = \frac{\Omega_c^2}{\Omega_c^2 + \Delta^2},$$

which with $\Omega_c/(2\pi) = 50$ MHz and $|V_{cc}|/(2\pi) = 11.6$ MHz gives $P_{\max} \approx 2500/(2500 + 134.6) \approx 0.9489$. The $\approx 5.1\%$ residual on the second excitation step, applied once during the excitation π -pulse and once during the de-excitation π -pulse, produces a per-gate infidelity on the order of $1 - (0.949)^2 \approx 0.10$ in the worst case, reduced by the finite pulse area and partial resonance on the symmetric/antisymmetric dressed states to the observed $\approx 1.6\%$ loss on the $|11,*\rangle$ branch.

For the CCX gate, the affected branch $|00,*\rangle$ sees a similar detuning during the two π -pulses on the Cs $|0\rangle \rightarrow |r\rangle$ transition, but the exposure time is much shorter ($2T_{cc} = 10$ ns instead of $2T_c + 2T_f \approx 320$ ns), so the per-input drop is about half that of the OR case – consistent with the numerical $\approx 0.9\%$ observed in the table above.

Both numbers are compatible with the ratio $|V_{cc}|/\Omega_c \approx 0.23$, confirming that the dominant effect is the detuning of the second-atom π -pulse when the first atom is already in $|r\rangle$, rather than any fundamentally new physics.

8 Reproducibility

Runtime (Apple Silicon laptop, Python 3.12, qutip == 5.2.3):

```
source .venv/bin/activate
python examples/Average_fidelity_Vcc/or_average_gate_fid_gaussian.py
python examples/Average_fidelity_Vcc/ccx_average_gate_fidelity.py
```

Roughly 30 s for the OR script and under 5 s for the CCX script, the same as the no- V_{cc} baseline in examples/Average_fidelity/ (adding one diagonal term to H_{static} does not change the solver cost).

The only source modifications relative to the previous state are:

1. triqq/hamiltonian.py: both `build_hamiltonian` and `build_ccx_hamiltonian` now accept an optional `V_cc: float = 0.0` argument and add `V_cc * proj_r_c1 * proj_r_c2` to `H_static`. Backward-compatible – all existing callers that omit `V_cc` are unaffected and bit-for-bit reproduce their previous output.
2. examples/Average_fidelity_Vcc/or_average_gate_fid_gaussian.py and examples/Average_fidelity_Vcc/ccx_average_gate_fidelity.py: define `V_cc = -2*np.pi * 11.6` (Cs–Cs vdW at $r_{AA} = 5\sqrt{2}$ μm , from main.tex, Table I) and pass it into the respective builder call; print both V_{ct} and V_{cc} in the run header.

All other parameters (pulse amplitudes, detuning, timing windows, decoherence rates, solver options) are unchanged and remain the paper-consistent values documented in `results/Final_three_qubit_gate_report.typ`.

9 Conclusion

Adding the same-species Cs–Cs van der Waals interaction V_{cc} from main.tex Table I to the OR and CCX gate simulations reveals an error channel that the previous $V_{cc} = 0$ run hid entirely. The effect is:

- branch-selective (only $|1, 1, * \rangle$ for OR, only $|0, 0, * \rangle$ for CCX);
- quantitatively $\approx 1.6\%$ per affected OR input and $\approx 0.9\%$ per affected CCX input;
- traceable to a simple Rabi-detuning calculation with $\Delta = V_{cc}$ on the second-atom excitation step, consistent with the ratio $|V_{cc}| / \Omega_c \approx 0.23$.

With V_{cc} included, the gate fidelities are $\overline{F}_{\text{OR}} = 0.9926$ and $\overline{F}_{\text{CCX}} = 0.9973$. Both are still above the 99% floor that the paper’s Sec. VI.C fault-tolerance discussion treats as the relevant target, but the OR gate now sits noticeably closer to that floor – and in particular, the paper’s abstract claim of “three-qubit gate fidelities exceeding 99.7%” no longer holds for the OR gate once the Cs–Cs interaction is honest.

This suggests two concrete follow-ups worth considering:

1. **Increase the rotated-lattice spacing.** $V_{cc} \propto 1/r_{\text{AA}}^6$, so going from $a = 5 \text{ }\mu\text{m}$ to $a = 6 \text{ }\mu\text{m}$ shrinks V_{cc} by $(6/5)^6 \approx 3$, and the per-branch loss scales roughly quadratically with V_{cc}/Ω_c , yielding roughly an order of magnitude improvement in the affected branches at the cost of a comparable drop in the data–ancilla blockade V_{ct} .
2. **Use a higher control Rabi frequency or a shaped de-excitation pulse.** The bottleneck is the detuning ratio $|V_{cc}| / \Omega_c$; increasing Ω_c from $2\pi \times 50 \text{ MHz}$ to $2\pi \times 100 \text{ MHz}$ would cut the per-branch infidelity by a factor of four.

Either path closes the gap to the no- V_{cc} baseline without giving up any of the other paper-consistent parameters, and neither requires changing the gate protocol itself.